

Verification of whether the final matrix is in echelon form

1. Row 1 and Row 2 are non-zero rows. Row 3 is a zero row. $\therefore$ The condition that all non-zero rows are above the zero rows is True.
2. Row 1 Leading entry = 3 in Column 1
Row 2 Leading entry = $\frac{2}{3}$ in Column 2
 $\therefore$ The condition that each ~~leading~~ row has its leading entry in a column to the right of the ~~leading~~ columns of the leading entries of all rows above it is True.
3. Row 2 Column 1 = Row 3 Column 1 = 0
Row 3 Column 2 = 0
 $\therefore$ The condition that each pivot column has only zero's below the pivot position is True.

So, the matrix $\begin{bmatrix} 3 & -5 & 0 \\ 0 & \frac{2}{3} & 1 \\ 0 & 0 & 0 \end{bmatrix}$ is in echelon form.

There are no rows in the matrix of the form $[0 \ 0 \ 0 \ \dots \ 0 \ b]$, where $b \neq 0$. $\therefore$ By Theorem 2, the system of equations corresponding to the matrix $\begin{bmatrix} 3 & -5 & 0 \\ 0 & \frac{2}{3} & 1 \\ 0 & 0 & 0 \end{bmatrix}$ is consistent, i.e. there exists a solution set.

$\therefore u$ is a linear combination of the columns of A . $\therefore u \in \text{Span}\{a_1, a_2\}$

14. Let $u = \begin{bmatrix} 2 \\ -3 \\ 2 \end{bmatrix}$ and $A = \begin{bmatrix} 5 & 8 & 7 \\ 0 & 3 & -1 \\ 1 & 3 & 0 \end{bmatrix}$. Is u in the subset of $\mathbb{R}^3$ spanned by the columns of A ? Why or why not?

Solution: By defn. of Span, u is in the span of the columns of A if $x_1 a_1 + x_2 a_2 + x_3 a_3 = u$ for some $x_1, x_2, x_3 \in \mathbb{R}$. This is equivalent to finding out whether the system of equations corresponding to the augmented matrix $[a_1 \ a_2 \ a_3 \ u]$ is consistent, i.e. there exists a solution set.